

Maximize

$$\begin{aligned} & 2x_{_000} + 0.8x_{_001} + 0.6x_{_002} + 2.4x_{_010} + 1.5x_{_011} + 1.5x_{_012} \\ & + 1.9x_{_020} + x_{_021} + 0.7x_{_022} + x_{_100} + 0.8x_{_101} - 0.4x_{_102} \\ & - 0.6x_{_110} - 0.5x_{_111} - 1.5x_{_112} + 1.9x_{_120} + 2x_{_121} + 0.7x_{_122} \\ & + 3x_{_200} + 2.8x_{_201} + 3.6x_{_202} + 0.4x_{_210} + 0.5x_{_211} + 1.5x_{_212} \\ & + 1.9x_{_220} + 2x_{_221} + 2.7x_{_222} + 2x_{_300} - 1.2x_{_301} - 0.4x_{_302} \\ & + 3.4x_{_310} + 0.5x_{_311} + 1.5x_{_312} + 4.9x_{_320} + 2x_{_321} + 2.7x_{_322} \end{aligned}$$

Subject To

$$\begin{aligned} \text{R0: } & 3x_{_000} + 3x_{_001} + 3x_{_002} + 3x_{_100} + 3x_{_101} + 3x_{_102} + 4x_{_200} \\ & + 4x_{_201} + 4x_{_202} + 4x_{_300} + 4x_{_301} + 4x_{_302} \leq 6000 \end{aligned}$$

$$\begin{aligned} \text{R1: } & 3.5x_{_010} + 3.5x_{_011} + 3.5x_{_012} + 3.5x_{_110} + 3.5x_{_111} + 3.5x_{_112} \\ & + 4.5x_{_210} + 4.5x_{_211} + 4.5x_{_212} + 4.5x_{_310} + 4.5x_{_311} + 4.5x_{_312} \\ & \leq 5000 \end{aligned}$$

$$\begin{aligned} \text{R2: } & 3x_{_020} + 3x_{_021} + 3x_{_022} + 3.5x_{_120} + 3.5x_{_121} + 3.5x_{_122} \\ & + 4x_{_220} + 4x_{_221} + 4x_{_222} + 4.5x_{_320} + 4.5x_{_321} + 4.5x_{_322} \\ & \leq 3000 \end{aligned}$$

$$\begin{aligned} \text{R3: } & 8x_{_000} + 8x_{_001} + 8x_{_002} + 8.5x_{_100} + 8.5x_{_101} + 8.5x_{_102} \\ & + 9x_{_200} + 9x_{_201} + 9x_{_202} + 9x_{_300} + 9x_{_301} + 9x_{_302} \leq 10000 \end{aligned}$$

$$\begin{aligned} \text{R4: } & 7x_{_010} + 7x_{_011} + 7x_{_012} + 7x_{_110} + 7x_{_111} + 7x_{_112} + 8x_{_210} \\ & + 8x_{_211} + 8x_{_212} + 9x_{_310} + 9x_{_311} + 9x_{_312} \leq 12500 \end{aligned}$$

$$\begin{aligned} \text{R5: } & 7.5x_{_020} + 7.5x_{_021} + 7.5x_{_022} + 7.5x_{_120} + 7.5x_{_121} + 7.5x_{_122} \\ & + 8.5x_{_220} + 8.5x_{_221} + 8.5x_{_222} + 8.5x_{_320} + 8.5x_{_321} + 8.5x_{_322} \\ & \leq 6000 \end{aligned}$$

$$\begin{aligned} \text{R6: } & x_{_000} + x_{_001} + x_{_002} + 1.1x_{_010} + 1.1x_{_011} + 1.1x_{_012} + 1.1x_{_020} \\ & + 1.1x_{_021} + 1.1x_{_022} + 1.1x_{_100} + 1.1x_{_101} + 1.1x_{_102} + x_{_110} \\ & + x_{_111} + x_{_112} + 1.1x_{_120} + 1.1x_{_121} + 1.1x_{_122} + 1.2x_{_200} \\ & + 1.2x_{_201} + 1.2x_{_202} + 1.1x_{_210} + 1.1x_{_211} + 1.1x_{_212} + 1.3x_{_220} \\ & + 1.3x_{_221} + 1.3x_{_222} + 1.3x_{_300} + 1.3x_{_301} + 1.3x_{_302} + 1.4x_{_310} \\ & + 1.4x_{_311} + 1.4x_{_312} + 1.3x_{_320} + 1.3x_{_321} + 1.3x_{_322} \leq 3500 \end{aligned}$$

$$\text{R7: } x_{_000} + x_{_010} + x_{_020} \leq 200$$

$$\text{R8: } x_{_001} + x_{_011} + x_{_021} \leq 400$$

$$\text{R9: } x_{_002} + x_{_012} + x_{_022} \leq 200$$

$$\text{R10: } x_{_100} + x_{_110} + x_{_120} \leq 300$$

$$\text{R11: } x_{_101} + x_{_111} + x_{_121} \leq 300$$

$$\text{R12: } x_{_102} + x_{_112} + x_{_122} \leq 400$$

$$\text{R13: } x_{_200} + x_{_210} + x_{_220} \leq 500$$

$$\text{R14: } x_{_201} + x_{_211} + x_{_221} \leq 200$$

$$\text{R15: } x_{_202} + x_{_212} + x_{_222} \leq 300$$

$$\text{R16: } x_{_300} + x_{_310} + x_{_320} \leq 200$$

$$\text{R17: } x_{_301} + x_{_311} + x_{_321} \leq 400$$

$$\text{R18: } x_{_302} + x_{_312} + x_{_322} \leq 300$$

$$\begin{aligned} \text{R19: } & x_{_000} + x_{_001} + 1.1x_{_010} + 1.1x_{_011} + 1.1x_{_100} + 1.1x_{_101} + x_{_110} \\ & + x_{_111} + 1.2x_{_200} + 1.2x_{_201} + 1.1x_{_210} + 1.1x_{_211} + 1.3x_{_300} \\ & + 1.3x_{_301} + 1.4x_{_310} + 1.4x_{_311} \leq 1500 \end{aligned}$$

I = {0, 1, 2, 3} 0: football size2, 1: football size3, 2: football size4, 3: football size5

J = {0, 1, 2} 0: production site1, 1: production site2, 2: production site3

K = {0, 1, 2} 0: RFEF, 1: TheFA, 2: FFF

$$\text{Maximize } Z = \sum_{i=0}^3 \sum_{j=0}^2 \sum_{k=0}^2 [(P_{ik} - C_{ij} - T_{jk}) \times x_{ijk}]$$

Maximize

$$\begin{aligned} & 2 x_{000} + 0.8 x_{001} + 0.6 x_{002} + 2.4 x_{010} + 1.5 x_{011} + 1.5 x_{012} + 1.9 x_{020} + \\ & x_{021} + 0.7 x_{022} + x_{100} + 0.8 x_{101} - 0.4 x_{102} - 0.6 x_{110} - 0.5 x_{111} - 1.5 x_{112} + \\ & 1.9 x_{120} + 2 x_{121} + 0.7 x_{122} + 3 x_{200} + 2.8 x_{201} + 3.6 x_{202} + 0.4 x_{210} + \\ & 0.5 x_{211} + 1.5 x_{212} + 1.9 x_{220} + 2 x_{221} + 2.7 x_{222} + \\ & 2 x_{300} - 1.2 x_{301} - 0.4 x_{302} + 3.4 x_{310} + 0.5 x_{311} + 1.5 x_{312} + 4.9 x_{320} + \\ & 2 x_{321} + 2.7 x_{322} \end{aligned}$$

Subject To

$$\sum_{i=0}^3 \sum_{k=0}^2 L_{ij} \times x_{ijk} \leq L_j^{max} \text{ for } j = 0,1,2$$

$$\text{R0: } 3 x_{000} + 3 x_{001} + 3 x_{002} + 3 x_{100} + 3 x_{101} + 3 x_{102} + 4 x_{200} + 4 x_{201} + 4 x_{202} + 4 x_{300} + 4 x_{301} + 4 x_{302} \leq 6000$$

$$\text{R1: } 3.5 x_{010} + 3.5 x_{011} + 3.5 x_{012} + 3.5 x_{110} + 3.5 x_{111} + 3.5 x_{112} + 4.5 x_{210} + 4.5 x_{211} + 4.5 x_{212} + 4.5 x_{310} + 4.5 x_{311} + 4.5 x_{312} \leq 5000$$

$$\text{R2: } 3 x_{020} + 3 x_{021} + 3 x_{022} + 3.5 x_{120} + 3.5 x_{121} + 3.5 x_{122} + 4 x_{220} + 4 x_{221} + 4 x_{222} + 4.5 x_{320} + 4.5 x_{321} + 4.5 x_{322} \leq 3000$$

$$\sum_{i=0}^3 \sum_{k=0}^2 M_{ij} \times x_{ijk} \leq M_j^{max} \text{ for } j = 0,1,2$$

$$\text{R3: } 8 x_{000} + 8 x_{001} + 8 x_{002} + 8.5 x_{100} + 8.5 x_{101} + 8.5 x_{102} + 9 x_{200} + 9 x_{201} + 9 x_{202} + 9 x_{300} + 9 x_{301} + 9 x_{302} \leq 10000$$

$$R4: 7x_{010} + 7x_{011} + 7x_{012} + 7x_{110} + 7x_{111} + 7x_{112} + 8x_{210} + 8x_{211} + 8x_{212} + 9x_{310} + 9x_{311} + 9x_{312} \leq 12500$$

$$R5: 7.5x_{020} + 7.5x_{021} + 7.5x_{022} + 7.5x_{120} + 7.5x_{121} + 7.5x_{122} + 8.5x_{220} + 8.5x_{221} + 8.5x_{222} + 8.5x_{320} + 8.5x_{321} + 8.5x_{322} \leq 6000$$

$$\sum_{i=0}^3 \sum_{j=0}^2 \sum_{k=0}^2 R_{ij} \times x_{ijk} \leq R^{max}$$

$$R6: x_{000} + x_{001} + x_{002} + 1.1x_{010} + 1.1x_{011} + 1.1x_{012} + 1.1x_{020} + 1.1x_{021} + 1.1x_{022} + 1.1x_{100} + 1.1x_{101} + 1.1x_{102} + x_{110} + x_{111} + x_{112} + 1.1x_{120} + 1.1x_{121} + 1.1x_{122} + 1.2x_{200} + 1.2x_{201} + 1.2x_{202} + 1.1x_{210} + 1.1x_{211} + 1.1x_{212} + 1.3x_{220} + 1.3x_{221} + 1.3x_{222} + 1.3x_{300} + 1.3x_{301} + 1.3x_{302} + 1.4x_{310} + 1.4x_{311} + 1.4x_{312} + 1.3x_{320} + 1.3x_{321} + 1.3x_{322} \leq 3500$$

$$\sum_{j=0}^2 x_{ijk} \leq D_{ik} \text{ for } i = 0,1,2,3 \text{ and } k = 0,1,2$$

$$R7: x_{000} + x_{010} + x_{020} \leq 200$$

$$R8: x_{001} + x_{011} + x_{021} \leq 400$$

$$R9: x_{002} + x_{012} + x_{022} \leq 200$$

$$R10: x_{100} + x_{110} + x_{120} \leq 300$$

$$R11: x_{101} + x_{111} + x_{121} \leq 300$$

$$R12: x_{102} + x_{112} + x_{122} \leq 400$$

$$R13: x_{200} + x_{210} + x_{220} \leq 500$$

$$R14: x_{201} + x_{211} + x_{221} \leq 200$$

$$R15: x_{202} + x_{212} + x_{222} \leq 300$$

$$R16: x_{300} + x_{310} + x_{320} \leq 200$$

$$R17: x_{301} + x_{311} + x_{321} \leq 400$$

$$R18: x_{302} + x_{312} + x_{322} \leq 300$$

$$\sum_{i=0}^3 \sum_{j=0}^1 \sum_{k=0}^1 x_{ijk} \leq I^{max}$$

$$\begin{aligned} R19: & x_{000} + x_{001} + 1.1 x_{010} + 1.1 x_{011} + 1.1 x_{100} + 1.1 x_{101} + x_{110} + x_{111} \\ & + 1.2 x_{200} + 1.2 x_{201} + 1.1 x_{210} + 1.1 x_{211} + 1.3 x_{300} + 1.3 x_{301} \\ & + 1.4 x_{310} + 1.4 x_{311} \leq 1500 \end{aligned}$$

$$x_{ijk} \geq 0$$